TROPIC Clinical Analysis Report

Study EFC6193 / XRP6258 -- Abbreviated Clinical Study Summary

SAP v4.0 lock note (2026-06-25): This report is a generated demonstration output under remediation control. It is not a clinical study report for submission and it is not a source of analysis requirements. The governing analysis plan is 02_specifications/sap/TROPIC_SAP_v4.0_industry_grade.docx.

Study Title: Prednisone plus Cabazitaxel or Mitoxantrone for mCRPC Progressing After Docetaxel

Sponsor: Sanofi-Aventis | Phase: III, Open-Label RCT | NCT: 00417079

Publication: de Bono JS et al., Lancet 2010;376(9747):1147-1154

Mixed real + synthetic data. All MP-arm statistics (N=371) are derived from the official Sanofi de-identified SDTM public data release (2013) -- real patient-level data. The Cabazitaxel (CbzP, N=378) arm is a synthetic, illustrative cohort built by two methods depending on endpoint: OS and PFS are reconstructed via genuine Guyot (2012) IPD reconstruction from the published Lancet 2010 Kaplan-Meier curves (independent of the MP arm, so the reconstructed HRs are non-circular and reproduce the published effect); the secondary time-to-event endpoints (TTPSA/TTUMOR/TTPAIN) are PH-scaled from the MP arm and are circular by construction; the non-TTE domains are fixed-seed sampled from published Lancet 2010 Table 1/Table 2 marginals. It is not real patient data and is shown only to exercise the analysis pipeline -- secondary CbzP-vs-MP comparisons are illustrative only, not clinical findings (see ADRG Section 7).

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1. Study Population

The Safety Population consisted of 371 patients in the Mitoxantrone + Prednisone (MP) arm, all with metastatic castration-resistant prostate cancer (mCRPC) progressing during or after docetaxel-based chemotherapy.

Characteristic	MP Arm (N=371)
ECOG Performance Status >=1	Majority
Prior docetaxel progression	100% (eligibility criterion)
Measurable disease	Subset (MEASDISF = Y)
Visceral disease	Subset (VISCFL = Y)
Baseline PSA (median)	~110 ng/mL
Baseline ALP (median)	~140 U/L

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2. Overall Survival & Progression-Free Survival -- Guyot Reconstruction (synthetic comparator)

The primary survival endpoints exercise the stratified Cox / log-rank machinery and the hierarchical step-down gatekeeping logic (ICH E9). For OS and PFS the synthetic CbzP arm is recovered by genuine Guyot (2012) IPD reconstruction (IPDfromKM) from the published de Bono 2010 Kaplan-Meier curves (Fig 2A = OS, Fig 3 = PFS) plus the transcribed numbers-at-risk tables. The survival shape comes from the published curve itself -- independently of the MP arm (no hazard-ratio scaling), so the CbzP-vs-MP hazard ratio is not circular: it emerges from the reconstructed CbzP curve versus the real MP data.

Statistic	**Synthetic CbzP (N=378, Guyot)(1)**	**Real MP (N=371)**	**Published (de Bono 2010)**
Median OS	15.2 months	12.7 months	15.1 mo vs 12.7 mo
Median PFS	2.7 months	1.4 months	2.8 mo vs 1.4 mo
OS HR (CbzP vs MP)	0.70 (95% CI 0.59-0.84)	Reference	0.70 (0.59-0.83)
PFS HR (CbzP vs MP)	0.72 (95% CI 0.62-0.84)	Reference	0.74 (0.64-0.86)

OS/PFS are reconstructed, not PH-scaled. Because the CbzP curve is inverted from the published figure rather than divided out of the MP arm, the reconstructed HRs are non-circular -- the OS HR matches the published 0.70 exactly and the PFS HR (0.72) is within tolerance of the published 0.74 (acceptance gates in 01_source_data/guyot_validation_report.md). The arm is still synthetic and illustrative -- not real patient data -- and is shown to exercise the pipeline, not as an independent clinical finding.

(1) Synthetic, illustrative -- not real patient data; reconstructed from the published KM curves (Guyot 2012).

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3. Secondary Time-to-Event & Response Endpoints -- Pipeline Demonstration (synthetic comparator)

The secondary time-to-event endpoints (TTPSA, TTUMOR, TTPAIN) use the synthetic CbzP arm PH-scaled from the real MP arm -- no published KM curves with at-risk tables exist for these, so Guyot reconstruction is not possible and their HRs/p-values are circular by construction (non-inferential). Response endpoints use the fixed-seed simulated CbzP cohort. Values are the live output of 05_outputs/tfl/tfl_generation.R (see 05_outputs/tfl/output/tables/T-11-Efficacy_Tables.txt).

Endpoint	**Synthetic CbzP(1)**	**Real MP**	**Pipeline HR / test**	**Pipeline p**	**Published (de Bono 2010)**
Time to PSA Progression(2)	2.7 mo (286/378)	2.2 mo (265/371)	HR 0.85 (0.72-1.00)	0.0514	median 6.4 mo, HR 0.75
Time to Tumour Progression(2)	9.1 mo (166/378)	2.4 mo (328/371)	descriptive KM	--	publication-era endpoint context
Time to Pain Progression(2)	7.9 mo (130/378)	NA (45/371)	HR 2.36 (1.68-3.32)	<0.0001	SAP T-11-8 mapping restored
PSA Response (>=50% decrease; baseline PSA >=20 ug/L)	40.2% (145/361)	18.5% (61/329)	Fisher's exact	5.2e-10	39% vs 24% (p = 0.0002)

Endpoint	Synthetic CbzP(1)	Real MP	Pipeline HR / test	Pipeline p	Published (de Bono 2010)
Overall Response Rate (ORR, confirmed)Section	16.8% (30/179)	6.4% (13/203)	Fisher's exact	0.0018	14.4% vs 4.4% (p = 0.005)
Pain Response (F-042)	N/A (PN unavailable)	28.1% (43/153)	descriptive responder rate	--	source-qualified PN/SV implementation

(1) Synthetic, illustrative -- not real patient data. (2) PH-scaled from the real MP arm where applicable; any HR is circular by construction (descriptive of synthetic data only, not a treatment-effect estimate). Section ORR is restricted to the measurable-disease subpopulation (CbzP N=179, MP N=203). TTUMOR is ITT-primary (CbzP N=378, MP N=371), with measurable disease retained as supportive. Pain response is unavailable for CbzP because no PN domain is reconstructed. The OS/PFS primary endpoints -- which are non-circular Guyot reconstructions -- are in Section 2.

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4. Safety -- Adverse Events (Safety Population)

Treatment-emergent adverse events (TEAEs) were defined as events occurring on or after the first dose date (TRTEMFL = "Y" in ADAE).

4.1 Overall TEAE Incidence

Category	CbzP (N=378)	%	MP (N=371)	%
Any TEAE	364	96%	328	88%
Any Grade >=3 TEAE	310	82%	147	40%
Any Serious TEAE (SAE)	145	38%	78	21%
Any TEAE leading to discontinuation	68	18%	0	0%

4.2 Grade >=3 TEAEs by System Organ Class (Top 6)

System Organ Class	CbzP (n, %)	MP (n, %)
Blood & Lymphatic System Disorders	293 (78%)	39 (11%)
Gastrointestinal Disorders	34 (9%)	6 (2%)
General Disorders & Admin Site Conditions	35 (9%)	36 (10%)

System Organ Class	CbzP (n, %)	MP (n, %)
Musculoskeletal & Connective Tissue Disorders	18 (5%)	35 (9%)
Infections & Infestations	8 (2%)	19 (5%)
Nervous System Disorders	4 (1%)	14 (4%)

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5. Laboratory Toxicity -- CTCAE Grade Shift

Baseline to worst post-baseline CTCAE grade shifts were derived from the ADLB dataset using BASEFL, ANL01FL, and ATOXGR variables.

5.1 ANC / Neutrophils -- Key Finding

Of 378 patients in the CbzP arm, 321 (84.9%) experienced Grade 3/4 neutropenia compared to 154 (41.5%) of 371 in the MP arm, highlighting the hematological toxicity signature of Cabazitaxel.

5.2 Haemoglobin

Grade 3/4 anemia occurred in 34 (9.0%) patients in the CbzP arm compared to 9 (2.4%) in the MP arm.

5.3 Platelets

Thrombocytopenia was rare, with Grade 3/4 thrombocytopenia occurring in 16 (4.2%) CbzP patients vs 5 (1.3%) MP patients.

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6. Exposure Analysis (ADEX)

Cycle-by-cycle exposure was captured in ADEX across parameters including:

* RDI (Relative Dose Intensity) -- ratio of actual to planned cumulative dose
* NCYCLE -- number of treatment cycles completed per patient
* CUMDOSE -- cumulative dose received
* NDELDOSE / NREDDOSE -- cycles with delays or dose reductions

FDA Project Optimus alignment: RDI in Cycle 1 was used as the E-R exposure proxy in the scatter plot analysis (F-17-1).

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7. Pipeline Technical Summary

Layer	Technology	Standard
Source Data	SAS7BDAT (Sanofi / Project Data Sphere)	CDISC SDTMIG v3.1.1 (trial-era)
ADaM Production	SAS 9.4	ADaMIG v1.3
Independent Validation	R 4.6.0 / Pharmaverse	ADaMIG v1.3
Reconciliation	diffdf package	100% cell-by-cell match
TFL Generation	ggplot2, survival, patchwork	ICH E3 / NEJM style
Orchestration	Python 3.10 (cibuild.py)	22-stage CI pipeline

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8. Data Provenance & Limitations

Real data (MP arm): All 371 MP-arm patients, 5,428 AE records, 266 OS events, and ~79,000 laboratory records are derived directly from the official Sanofi de-identified public SDTM release (dated June 2013).

Synthetic comparator (CbzP arm): The Cabazitaxel arm was not included in the Sanofi public data release. The CbzP arm used in figures and comparative tables is synthetic and illustrative, built by two methods depending on endpoint:

- OS and PFS are reconstructed via genuine Guyot (2012) IPD reconstruction (IPDfromKM) from the published de Bono 2010 KM curves (Fig 2A = OS, Fig 3 = PFS) + transcribed numbers-at-risk tables. The shape comes from the published curve independently of the MP arm, so the reconstructed OS/PFS HRs are non-circular and reproduce the published effect (OS HR 0.70 exactly; PFS 0.72 vs published 0.74). Validated against the published summary statistics in 01_source_data/guyot_validation_report.md.

- Secondary TTE endpoints (TTPSA, TTUMOR, TTPAIN) remain PH-scaled from the real MP arm (no published KM curves exist for them) and are circular by construction.

- Non-TTE domains (AE, laboratory, exposure, demographics) are fixed-seed sampled from published Lancet 2010 Table 1/Table 2 marginal distributions.

The arm is not real patient data; secondary CbzP-vs-MP comparisons are illustrative only. See ADRG Section 7 for the full reconstruction methodology.

Single source of truth. Every count, percentage, median, HR and p-value in this report is produced by 05_outputs/tfl/tfl_generation.R and written to 05_outputs/tfl/output/tables/.txt. Narrative numbers are transcribed from those files; the generated tables govern in case of any discrepancy.

Reference

de Bono JS, Oudard S, Ozguroglu M, et al. Prednisone plus cabazitaxel or mitoxantrone for metastatic castration-resistant prostate cancer progressing after docetaxel treatment: a randomised open-label trial. Lancet. 2010;376(9747):1147-1154.

* Local Copy: de_bono_lancet_2010.pdf